

Studio di funzione - Esercizi

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$$\text{N° 2} \quad y = \frac{x^2 - 6x + 8}{x - 3}$$

1. DOMINIO

$$x - 3 \neq 0 \rightarrow x \neq 3$$

$$\begin{aligned} D &= \mathbb{R} - \{3\} \\ &= (-\infty, 3) \cup (3, +\infty) \end{aligned}$$

2. SIMMETRIE

$$\text{PARI} \quad f(-x) = f(x)$$

$$\frac{(-x)^2 - 6(-x) + 8}{(-x) - 3} \stackrel{?}{=} \frac{x^2 - 6x + 8}{x - 3} \quad \text{NO!}$$

$$\text{DISPARI} \quad f(-x) = -f(x)$$

$$\frac{(-x)^2 - 6(-x) + 8}{(-x) - 3} \stackrel{?}{=} - \left(\frac{x^2 - 6x + 8}{x - 3} \right)$$

$$\frac{x^2 + 6x + 8}{-x - 3} \stackrel{?}{=} \frac{-x^2 + 6x - 8}{x - 3} \quad \text{NO!}$$

3. INTERSEZIONI

$$\begin{cases} y = \frac{x^2 - 6x + 8}{x - 3} \\ x = 0 \end{cases} \rightarrow \begin{cases} y = 8/-3 \\ x = 0 \end{cases} \rightarrow \begin{cases} y = -8/3 \\ x = 0 \end{cases}$$

$$\begin{cases} \frac{x^2 - 6x + 8}{x - 3} = 0 \\ y = 0 \end{cases} \rightarrow \begin{cases} x^2 - 6x + 8 = 0 \\ y = 0 \end{cases} \rightarrow x_{1,2} = \frac{6 \pm \sqrt{6^2 - 4 \cdot 1 \cdot 8}}{2} = \frac{6 \pm 2}{2}$$

$$(2, 0) \quad (4, 0)$$

4. SEGNO

$$N \quad \frac{x^2 - 6x + 8}{x - 3} > 0$$

$$N \quad x^2 - 6x + 8 > 0$$

$$\rightarrow \begin{cases} x > 2 \\ x > 4 \end{cases} =$$

$$D \quad x-3 < 0$$

$$D \quad x-3 > 0 \rightarrow x > 3$$

$$I_P = (2, 3) \cup (4, +\infty)$$

$$I_N = (-\infty, 2) \cup (3, 4)$$

5. ASINTOTI

$$D = (-\infty, 3) \cup (3, +\infty)$$

ASINTOTI
ORIZZONTALI

$$\Rightarrow \lim_{x \rightarrow \pm \infty} f(x) = m \rightarrow \text{NUMERO FINITO}$$

ASINTOTI
VERTICALI

$$\Rightarrow \lim_{x \rightarrow m^+} f(x) = \pm \infty$$

ASINTOTI
OBBLIQUE

$$\Rightarrow y = mx + q$$

SE SONO VALORI FINITI

$$m = \lim_{x \rightarrow \pm \infty} \frac{f(x)}{x}$$

$$q = \lim_{x \rightarrow \pm \infty} [f(x) - mx]$$

$$\lim_{x \rightarrow \pm \infty} \frac{x^2 - 6x + 8}{x - 3} = \lim_{x \rightarrow \pm \infty} \frac{x^2(1 - \frac{6}{x} + \frac{8}{x^2})}{x(1 - \frac{3}{x})} = \pm \infty$$

$$\lim_{x \rightarrow 3^-} \frac{x^2 - 6x + 8}{x - 3} = \frac{9 - 18 + 8}{0^-} = \frac{-1}{0^-} = +\infty \quad \text{ASINTOTI VERTICE}$$

$$\lim_{x \rightarrow 3^+} \frac{x^2 - 6x + 8}{x - 3} = \frac{9 - 18 + 8}{0^+} = \frac{-1}{0^+} = -\infty \quad \text{ASINTOTI VERTICE}$$

$$m = \lim_{x \rightarrow \pm \infty} \frac{x^2 - 6x + 8}{x} = \lim_{x \rightarrow \pm \infty} \frac{x^2 - 6x + 8}{x - 3} \cdot \frac{1}{x} =$$

$$= \lim_{x \rightarrow \pm \infty} \frac{x^2 - 6x + 8}{x^2 - 3x} = \lim_{x \rightarrow \pm \infty} \frac{x^2(1 - \frac{6}{x} + \frac{8}{x^2})}{x^2(1 - \frac{3}{x})} =$$

1.

- 2

$$q = \lim_{x \rightarrow \pm \infty}$$

$$\boxed{y = mx + q} \rightarrow y = x - 3$$

X	Y
0	-3
1	-2
3	0

6. DISEGNO



